East West University
Department of Computer Science and Engineering
Course Outline
Spring 2026 Semester

Course Information

Course: CSE251/ICE213 Electronic Circuits (Sections: 3 & 4)

Credits and Teaching Scheme

	Theory	Laboratory	Total
Credits	3	1	4
Contact Hours	3 Hours/Week for 13 Weeks + Final Exam in the 14th Week	2 Hours/Week for 13 Weeks	5 Hours/Week for 13 Weeks + Final Exam in the 14th Week

Prerequisite

CSE209 Electrical Circuits

Instructor Information

Instructor: **Dr. Sarwar Jahan**
Associate Professor, Department of Computer Science and Engineering,
East West University, Bangladesh.
Office: Room # 444
Tel. No.: 165 (ext.)
E-mail: sjahan@ewubd.edu

Class Routine and Office Hour

Day/Time	08:00-08:30	08:30-10:00	10:10-11:40	11:50-01:20	01:30-03:00
Sunday	Office Hours	Office Hours	CSE251/ICE213 Section: 3 Room: FUB-401	CSE251/ICE213 Section: 4 Room: FUB-403	
Monday	Office Hours	CSE209/ICE109 Section: 1 Room: 217	CSE251/ICE213[LAB] Section: 4 Room: 449 Time: 10:10-12:10	Office Hours Time: 12:10-02:10	
Tuesday	Office Hours	Office Hours	CSE251/ICE213 Section: 3 Room: FUB-401	CSE251/ICE213 Section: 4 Room: FUB-403	CSE209/ICE109[LAB] Section: 1 Room: 547 Time: 01:30-03:30
Wednesday	Office Hours	CSE209/ICE109 Section: 1 Room: 217			
Thursday	Office Hours	Office Hours	CSE251/ICE213 [LAB] Section: 3 Room: 449 Time: 10:10-12:10		

Course Objective

The subject aims to provide students with the ability to use fundamental electrical and electronic abstractions to analyze and design circuits and systems built with lumped and electronic circuit elements. This course provides basic knowledge of how complex devices such as semiconductor diodes, operational amplifiers (op-amps), and bipolar and field-effect transistors are modeled and used to design and analyze practical circuits. Besides, this course also emphasizes the practical implementation of building, testing, and performance analysis of electronic circuits. Knowledge of this course will be a prerequisite for future courses such as CSE345 Digital Logic Design, CSE350 Data Communications, CSE360 Computer Architecture, CSE442 Microprocessor and Microcontrollers, and CSE490 VLSI Design.

Knowledge Profile

K1: Theory-based natural sciences
K3: Theory-based engineering fundamentals

Learning Domains

Cognitive - C2: Understanding, C3: Applying, C4: Analyzing
Psychomotor - P2: Manipulation, P3: Precision
Affective - A2: Responding

Program Outcomes (POs)

PO1: Engineering Knowledge
PO2: Problem Analysis

Complex Engineering Problem Solution

EP1: Depth of knowledge required
EP2: Range of conflicting requirements

Complex Engineering Activities

None

Course Outcomes (COs) with Mappings

After completion of this course, students will be able to:

CO	CO Description	PO	Learning Domains	Knowledge Profile	EP/EA
CO1	**Understand** and **use** the fundamental concepts of Diode, Bipolar Junction Transistor (BJT), MOSFET, and Operational Amplifier (Op-amp) for solving electronic circuits.	PO1	C2, C3	K1	
CO2	**Use** and **justify** Diode, BJT, and MOSFET for designing electronic circuits.	PO2	C3	K2	
CO3	**Use** and **justify** Operational amplifiers for designing electronic circuits.	PO2	C3	K2	

CO4	**Use** analytical, software, and hardware techniques, **perform** and **demonstrate** skills, and **write** reports to design, build and test electronic circuits.	PO1	C3 P2, P3 A2	K1, K2	EP1, EP2

Course Topics, Teaching-Learning Method, and Assessment Scheme

Lecture Plan (L)	Topic	Teaching-Learning Method	CO	Mark of Cognitive Learning Levels			Mark of COs	Exam Mark
				C2	**C3**	**C4**		
L1-L2	Operation and characteristics of semiconductor diode, Load-line analysis.	Lecture, Slides, Class Discussion, Discussion with Instructor/TA	CO1	3	4		7	Mid-Semester Assessment (30)
L3-L6	Applications of Diode: Rectifier circuits, Clipper and Clamper, Zener Diode.	Do	CO2		10		10	
L7-L9	Device Structure and Physical Operation of BJT, Modes of Operation, Current-Voltage Characteristics, BJT as an amplifier and switch.	Do	CO1	5			5	
L10-L12 [Quiz# L11]	BJT circuits at DC, Biasing in BJT amplifier circuits.	Do	CO2			8	8	
L13-L15	Characteristics of the ideal Op-amp, Basic Comparator circuits, Inverting and non-inverting amplifiers, Voltage follower.	Do	CO1	3	4		7	Final Exam (30)

L16-L18 [Quiz# L18]	Applications such as Adder and Difference amplifiers, Integrator and Differentiator, and Instrumentation amplifiers.	Do	CO3		3	5	8	
L19-L20	Device Structure and Physical Operation of MOSFET, Modes of Operation, Current-Voltage Characteristics, MOSFET as an Amplifier and Switch.	Do	CO1	3	4		7	
L21-L24 [Quiz# L22]	DC biasing and small signal operations of MOSFET, Small signal equivalent models of MOSFET, Logic circuit using n-MOS and p-MOS.	Do	CO2		3	5	8	

Laboratory Experiments and Assessment Scheme

Experiment	**Teaching-Learning Method**	**CO**	**Mark of Cognitive Learning Level**	**Mark of Psychomotor Learning Levels**		**Mark of Affective Learning Level**	**CO Mark**
			C3	**P2**	**P3**	**A2**	
I-V Characteristics and Modeling of Forward Conduction of a Diode	Preparing Pre-Lab Report, Lab Experiments, and Result Analysis	CO4					
Diode Rectifier Circuits	Do	CO4					
Study of clipper and clamper circuits	DO	CO4					

Biasing of BJT Common-Emitter Circuit	Do	CO4					
Adder and Amplifier Circuits Using Op Amp	Do	CO4					
Signal Integration and Differentiation Using Op-Amp	Do	CO4					
Measurement of Parameters and I-V characteristics of an N-channel MOSFET	Do	CO4					
Biasing of a Common-Source Voltage Amplifier	Do	CO4					
Total Lab Performance		CO4	4	1	0	0	5
Lab Exam		CO4	7	1	1	1	10
Total			**11**	**2**	**1**	**1**	**15**

Mini Project

Mini Project	Teaching-Learning Method	CO	EP/ EA	Mark of Cognitive Learning Level	Mark of Psychomotor Learning Levels		Mark of Affective Learning Level	CO Mark
				C3	**P2**	**P3**	**A2**	

Mini Project including Report and Presentation	Group-based, moderately complex electronic circuit building for practical application with report writing and presentation	CO4	EP1, EP2	2	1	1	1	5

Overall Assessment Scheme

Assessment Area	**CO**					**Total**	**PO Marks**	
	CO1	**CO2**	**CO3**	**CO4**	**Other**		**PO1**	**PO2**
Class Test/Quiz	5	5	5			15	5	10
Mid-Semester Assessment	12	18				30	12	18
Final Exam	14	8	8			30	14	16
Laboratory Performance and Lab Exam				15		15	15	
Mini Project				5		5	5	
Class Performance					5	5		
Total	**31**	**31**	**13**	**20**	**5**	**100**	**51**	**44**

Teaching Materials

Textbook:

1. Adel S. Sedra and Kenneth C. Smith, *Microelectronic Circuits*, New Delhi, Oxford University Press, 2010.
2. Charles K. Alexander and Matthew N. O. Sadiku, *Fundamental of Electrical Circuits*.New Delhi, McGraw Hill Education, 2013.
3. Robert Boylestad, and Louis Nashelsky, Electronic devices, and circuit theory, Pearson, 2012.

Lab Manual: Lab manuals will be provided.

Project Description: Project descriptions will be provided.

Equipment/Software: Digital trainer board, and Computer & PSpice Software.

Grading System

Marks (%)	Letter Grade	Grade Point
80% and above	A+	4.00
75% to less than 80%	A	3.75
70% to less than 75%	A-	3.50
65% to less than 70%	B+	3.25
60% to less than 65%	B	3.00
55% to less than 60%	B-	2.75
50% to less than 55%	C+	2.50
45% to less than 50%	C	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00

Exam Dates

Section	Class Slot	Mid Semester	Final
3&4	ST	01 March 2025 (Sunday)	26 April 2026 (Sunday)
**As per the university decision, Exam Dates may change.			

Academic Code of Conduct

Academic Integrity:
Any form of cheating, plagiarism, personification, or falsification of a document, as well as any other form of dishonest behavior related to obtaining academic gain or the avoidance of evaluative exercises committed by a student, is an academic offense under the Academic Code of Conduct and **may lead to severe penalties as decided by the Disciplinary Committee of the university.**

Special Instructions:

- Students are expected to attend all classes and examinations. A student MUST have at least 80% class attendance to sit for the final exam.
- Students will not be allowed to enter the classroom after 20 minutes of the starting time.
- For plagiarism, the grade will automatically become zero for that exam/assignment.
- Normally there will be **NO make-up exam**. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student misses any exam, the student MUST get approval for a makeup exam by written application to the Chairperson through the Course Instructor **within 48hours**of the exam time. Proper supporting documents in favor of the reason for missing the exam have to be presented with the application.
- For the **final exam**, there will be NO makeup exam. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student misses the final exam, the student MUST get the approval of **Incomplete Grade** by written application to the Chairperson through the Course Instructor **within 48 hours** of the final exam time. Proper supporting documents in favor of the reason for missing the final exam have to be presented with the application. **It is the responsibility of the student to arrange an**

Incomplete Exam within the deadline mentioned in the Academic Calendar in consultation with the Course Instructor.

- All mobile phones MUST be turned to silent mode during class and exam periods.
- There is **zero tolerance for cheating** in exams. Students caught with cheat sheets in their possession, whether used or not; writing on the palm of the hand, back of calculators, chairs, or nearby walls; copying from cheat sheets or other cheat sources; copying from other examinees, etc. would be treated as cheating in the exam hall. The only penalty for cheating is **expulsion for several semesters as decided by the Disciplinary Committee of the university**.

Course Instructor